Vantage® System Release Notes

VANTAGE SYSTEM RELEASE 4.9.7

Verasonics Inc. | 11335 NE 122nd Way, Ste 100, Kirkland, WA 98034 | www.verasonics.com

Summary

This document summarizes the changes and new features included in the Vantage 4.9.7 system release and provides a summary of known issues to be addressed in future releases. The Vantage System release 4.9.7 contains the entire feature set for the 4.9 release and is an all-customer release. Vantage System releases 4.9.3, 4.9.4, and 4.9.5 were internal development or select customer releases.
For reference, the release notes from the previous Vantage System releases (4.9.6) are included in the documentation folder of your Vantage release.

Table of Contents

1. MATLAB® and Host Computer OS Compatibility

The Vantage System release 4.9.7 has been tested for compatibility and full functionality with the versions of MATLAB and host computer Operating Systems as listed in the following sections.

1.1. MATLAB

Verasonics recommends using MATLAB release 2022a, but any MATLAB release from 2020b through 2022a can be used. The 4.9.7 release will not function and will report a fault condition if a MATLAB release older than 2020b is detected. To avoid these problems, simply update your MATLAB installation to 2022a, 2021b, 2021a, or 2020b.

For all Vantage software installations, the MATLAB Signal Processing Toolbox is also required. Some of the "specialty applications" included with the Vantage software require the installation of additional MATLAB toolboxes, as explained in the documentation for each of those applications.

1.2. WINDOWS OS

Windows 10 is required for use with the Vantage software release 4.9.7 for all system configurations, along with version 12.9.0 of the driver for the hardware system. The driver installation instructions can be found in the *Vantage System User Manual*.

NOTE: If you are using the computer in software-only mode with no hardware system connected, changing the driver will not be required when switching between different Vantage software releases.

1.3. macOS®

Verasonics does not support the use of the Vantage System release 4.9.7 with a hardware system under macOS. For a software-only installation, any macOS version from Yosemite (10.10) or newer can be used.

1.4. Linux

For all system configurations using the 4.9.7 software, Ubuntu Version 18.04 LTS is required along with a Linux kernel version in the range 3.19 to 5.18.2, and the installation of driver version 15.0.0 for use with the hardware system.

For software-only installations on a computer that will never be used with a hardware system, installation of the driver is not required.

Accessing Documentation through the Verasonics Community Web Portal

Verasonics provides a growing list of Application Notes and other documentation for our customers through our community web portal. This website is restricted to Verasonics customers and requires a one-time registration. It can be reached by browsing the following URL: https://verasonicscommunity.com/. After signing up and entering the site, please click the Vantage Resource Documents tab to access our list of Application Notes, Manuals, and other helpful documentation about the Vantage System.

2. Feature Updates in the Vantage System Release 4.9.7

For all new features, refer to the Verasonics Community web portal for specific application notes. (Note: Not all release features have an application note in the Verasonics Community portal.)

3. Defects Corrected in the Vantage System Release 4.9.7

VSX: Spectrum Viewer (script SetUpL11_5vAcquireRFSigSpctrmViewer.m) no longer errors when changing frame number (Issue 9154)

In release 4.9.7, the spectrum viewer no longer errors when changing the frame number.

Row-Column Array script no longer crashes in recon for reconstruction pixels outside the apertures. (Issue 9403)

In release 4.9.7, Matlab will no longer crash when adding pixels outside the footprint of the row-column array. This can be achieved by setting the variable scale to either 1 or 2 when using a RCA script.

4. Known Issues in the Vantage System Release 4.9.7

FUS Elite 3000 3D: VSX prompts to reflash the SHI FPGAs on Initial Startup after installation of new Vantage Software release version.

After installing a new Vantage System version (for example, 4.9.7 to 4.9.2), if you run FUS Elite 3000 3D immediately after activating it in the new installation folder and launching the GUI program by typing `startFUSElite()` at the MATLAB Command prompt, VSX prompts to reflash SHI FPGAs after pressing Start on Initial Setup window. (Issue 5703)

When a new Vantage Software release is installed, ensure that you check for FPGA updates using the steps below:

1. Launch MATLAB and navigate to the Vantage Project folder.
2. In the MATLAB Command Window, type activate, and press ENTER.
3. Type the `reprogramHardware` command and press Enter. Pay attention to all the prompts and status updates displayed in the MATLAB Command Window.
4. Note that the `reprogramHardware` command forces revision checks and updates to all FPGA code files in the Vantage hardware system and this process may take several minutes to complete.
   Note: Do not interrupt the system while the update process is in progress.
5. Shut down the host controller to a full power-off state—a full power cycle is required for the FPGA file updates to take effect.
6. Shut down the Vantage hardware System.
7. First power on the Vantage hardware System and then power on the host controller.

FUS Elite 3000 3D: Allow reference frame to be taken for negative values of theta

Since the Vantage software release version 4.8.4, in FUS Elite3000 3D, reference frames and reference scans are limited to positive position values, between 0-179.9 degrees, in the Motion Control. (Issue 5182)

FUS Elite 3000 3D: Taking a reference scan with a one-degree step size using either Doppler imaging mode throws runAcq error

Since the Vantage software release version 4.8.4, in FUS Elite 3000 3D, acquiring a reference scan while in Doppler imaging mode within a one-degree step size will produce a RunAcq error due to insufficient memory (if the host controller has less than 16 GB of RAM). The user will need to use a different step size while in Doppler imaging for a reference scan. (Issue 5171)

FUS Elite 3000 3D: Doppler color bar displayed for non-Doppler scripts after loading presets saved in Doppler imaging mode

Since the Vantage software release version 4.8.4, in FUS Elite 3000 3D, the Doppler color bar may incorrectly display in non-Doppler mode after loading presets that were saved in Doppler mode. (Issue 5168)

FUS Elite 3000 3D: HIFU Treatment mode: RunAcq error in TSI mode if pre-scan and post-scan is disabled

Since the Vantage software release version 4.8.4, in FUS Elite 3000 3D, if pre-scan and post-scan are turned off on a reference frame or scan using TSI imaging type in HIFU treatment operating mode, a RunAcq error occurs. A workaround is to keep both pre-scan and post-scan selected during TSI imaging while in HIFU treatment operating mode. (Issue 5137)

FUS Elite 3000 3D: Calculate Burst PRF based on Image-Burst Interval and Imaging Mode TTNEB errors

Since the Vantage software release version 4.8.4, in FUS Elite3000 3D, while performing a HIFU Treatment a user is notified of TTNEB (time to next extended burst**)** warnings with higher Burst pulse repetition frequencies (Burst PRF). This is due to the acquisition data for imaging between bursts taking longer than the interval time between HIFU bursts. For higher Burst PRF frequencies, it may require some trial and error to dial in the settings necessary to achieve the desired result. (Issue 4940, Issue 4954)

FUS Elite 3000 3D: Review State cannot be stopped right now

Since the Vantage software release version 4.8.4, in FUS Elite 3000 3D, while using the Review state of HIFU Treatment the user is unable to stop the animation of the HIFU Treatment review. (Issue 4717)

FUS Elite 3000 3D: Data Tip in XZ plots reads "[X,Y]"

In all Vantage System releases since 4.8.4 in FUS Elite 3000 3D, the XZ plots in both Ref Data Review and HIFU Treatment read [X, Y] but are correct to be [X, Z] coordinates. (Issue 4889)

FUS Elite 3000 3D: Homing motor timeout causes GUI to freeze

The GUI can become unresponsive if the motor homing process results in a timeout. The issue is mitigated by using the micro-USB cables provided with the system as the timeouts can be more frequent when using different micro-USB cables. (Issue 6625)

Saving cine loop fails with spaces in the path

Saving a cine loop will fail with spaces in the path when using the Verasonics display window. As a workaround, save the file to a path that does not contain spaces. If saving to the *Verasonics Project Folder*, install the *Verasonics Project Folder* on a path without spaces. (Issue 7925)

FUS 2D System Test function sonicates without confirmation

The FUS 2D SystemTest button will run a live HIFU therapy plan without first confirming that the user is ready to run live therapy. (Issue 7231)

QuickScan: Cannot delete saved frames or cine loops from within GUI

Right-clicking on captured frames or cine loops and clicking “Delete” from the context menu does not delete the saved data when attempting to delete a frame other than the most recently captured frame. (Issue 6286)

QuickScan: Saving annotations behavior lacks clarity

In QuickScan, users need to manually select “Capture/Save” when adding frame annotations to a single frame or cineloop frame. Annotations are not automatically saved on the frame or cineloop frame. (Issue 6285)

System Malfunction with Incorrect Receive Buffer rowsPerFrame

All receive acquisition events are automatically rounded up if needed by the system software, to an integer multiple of 128 samples. As a result, the `rowsPerFrame` value used to define the receive buffer should also be a multiple of 128, and large enough to accommodate the largest receive buffer to be used while the sequence is running (the size of the receive buffer cannot be changed after the sequence has started). However, we have found that if you mistakenly set `rowsPerFrame` to a value that is not a multiple of 16, the system will run with no error reported and yet a meaningless corrupted image will be produced. To avoid this situation, simply make sure you specify Receive Buffer `rowsPerFrame` as a multiple of 128. Error reporting for this mistake will be added in a future release. (Issue 2630)

NDE GUI Linear Range Colormap values don't match the display

When using the Linear Display Resolution selection, if you adjust the Image Control Linear Range slider to its maximum value of 100, the actual image display will not match the identified color map. This will be corrected in a future release. (Issue 2486)

MATLAB Crash after Hardware System Fault

If the Vantage hardware system experiences a fault condition while a sequence is running—such as system overheating, transmit power supply over-voltage, or overcurrent fault—the system will automatically terminate execution and report the fault.

In some cases, a MATLAB crash may occur as a side effect of the forced termination of the script. Usually, you will still be able to see the error message at the MATLAB command prompt identifying the fault that occurred, but in some cases, the MATLAB crash report may cover it. If you encounter this situation, you will need to quit and restart MATLAB to recover. Note that for some hardware system faults, the system will put itself in a "locked down" state and you will have to completely shut down the system to a full power-off state and then restart it to restore normal operation. After a hardware system fault has occurred, it is a good idea to run the "VVT" system self-test to make sure the system is still fully functional. (Issues 2462, 2470)

QuickScan Anomalies

Since the 4.5.3 release, there are some known minor defects with the QuickScan feature: Changing probes while QuickScan is running may lead to an error condition; to avoid this, close QuickScan and restart it after changing probes (issue 2492).

While running an imaging script with QuickScan, if you switch to running a color Doppler script the color region in the image display may be incorrect; to avoid this, do not switch to the color Doppler mode while running QuickScan. It is recommended that you close and restart the QuickScan application to run the color Doppler mode. These issues will be corrected in a future release. (Issue 2491)

Abnormal exit from user-script leaves the hardware system running

This bug has been identified in releases 4.1.0 up through the current release: In some situations, while a user script is running with the hardware system, an event may occur that causes the Vantage software to close without triggering any error conditions. This situation can result in the hardware system continuing to run even though MATLAB has returned to the command prompt as if no sequence was running and with no errors or warnings reported. One way to get into this situation is by entering "control-c" at the MATLAB command prompt, while an asynchronous script is running this will cause VSX to close and return to the MATLAB prompt, but without closing the Hal or stopping the hardware system event sequence. You can tell this has occurred by observing the status indicators on the front of the Vantage System- they will remain on indicating the hardware event sequence is still running). Synchronous acquisition scripts are not affected by this problem (for example, those using a "sync" Sequence Control command or the "wait for processing" condition in Transfer To Host commands).

If you encounter this situation while running one of the affected releases, you can force the hardware system to stop by simply closing MATLAB (within 5 to 10 seconds after that, the software watchdog feature will stop the hardware sequence and the front panel LEDs will all go off). Then simply restart

MATLAB, select the Vantage software folder, and run activate. This will re-initialize the system and restore normal operation; there is no need to go through a full power-off cycle.

If the Vantage System is being used for clinical use with live subjects, the customer is responsible for completing a risk assessment according to their standards and regulations. Reference: Vantage Risk Analysis Revision E July 29, 2020, Hazard ID: H.AP.4

This problem will be corrected in a future software release. (Issue 1620)

The Microvascular Imaging example script has been removed

The Microvascular example in the *Specialty_Applications* folder has been removed in the 4.5.3 release, due to known bugs that prevented it from functioning properly. A new Microvascular example with improved performance is in development and will be added in a future release. (Issue 2184)

Receive Data Interleave Processing Malfunction

This bug affects the Recon pre-processing function for combining interleaved RF data acquisitions to create a higher RF data sample rate. Very intermittently and on only a few channels at a time, the interleave processing may not have been completed on all threads, and thus the RF data near the end of the frame will still be in the interleaved format. This effect is more likely to occur on more complex acquisition sequences with a large number of acquisitions per frame. If you encounter this problem a workaround that should eliminate it is to set `Receive.startDepth` to zero for all the interleaved acquisitions; a full solution will be incorporated in a future Vantage release. (Issue 638)

macOS Driver No Longer Available

macOS releases beyond 10.10 Yosemite have new driver requirements that prevent the driver from functioning properly. A new version of the driver may be provided in a future release. macOS may be used in simulation mode provided it complies with the compatible restrictions described in the Host OS Compatibility section of this document.

Receive A/D Sample Rate Initialization

The receive A/D rate of the system is set by VSX during initialization, as redefined by the `Receive` structures in the event sequence. The A/D sample rate clock signal is also used to control timing within hardware acquisition events, including synchronization to an external trigger input signal. In this and all earlier Vantage releases, the actual A/D sample rate being used by the hardware system is not initialized until the first Receive event in the event sequence is executed. This means the execution time for that first Receive event will be longer than usual to complete the sample rate initialization, and any timing activity prior to that event (such as responding to a trigger input signal) may also be different than in subsequent events while the script is running.

If the timing anomalies are unacceptable for the intended use of a script, a simple workaround is to place a dummy Receive event at the start of the script prior to any actual acquisition events or timing-related sequence control commands. This workaround may be particularly important to “single-shot” scripts that only execute once and do not repeat. (Issue 117)

Receive Mode 2 (accumulate from a reference acquisition)

The receive mode 2 accumulate feature is intended to allow you to acquire and save in hardware system memory a "reference acquisition" that can subsequently be combined with real-time acquisition events. This feature is not functional in the current release; it will be restored (in a more user-friendly implementation) in a future release. (Issue 202)

Triple Mode Example Script Does Not Function Properly

The "TripleModeSimultaneous" example script from earlier Vantage software releases does not function properly in the current software release. It has been removed and will be restored in a future release after it has been updated for full compatibility with the current system software. (Issue 144)

Intermittent MATLAB Crash in Simulation

In verification testing of the current release, intermittent MATLAB crashes were identified when running a very large number of scripts in simulation mode (without connecting the vantage hardware system). This condition is very difficult to reproduce and does not always affect the same script or even similar types of scripts but does appear to be directly tied to the use of the transmit-receive simulation function. Another situation that can lead to a MATLAB crash while running in simulation mode (MacOS) is when using a TX.Apod array consisting of only zeros. When one of these simulation-mode crashes occurs, it usually can be cleared by quitting the MATLAB application, restarting it, and then running the same script again. (Issue 412, 8953)